

Contents:

1.0	INTRODUCTION AND PROJECT OVERVIEW	2
2.0	STRUCTURAL DESIGN BRIEF	3-4
3.0	HYDRAULICS DESIGN BRIEF	
3.1	Drainage study	5-6
3.2	Flood study	6-7
4.0	GEOTECHNICAL DESIGN BRIEF	8-9
5.0	GENERAL SUBMISSION REQUIREMENTS	9

1.0 INTRODUCTION & PROJECT OVERVIEW:

A brownfield site on the edge of a city centre, indicated by the greyed area in Figure 1, is to have two new multi-storey buildings, A and B. The site is intersected by a river which sits within reinforced concrete training walls. Both the north and south half of the site were previously occupied by an extensive railway goods yard and maintenance depot which was in place for 100 years until the 1970s. The site was partially redeveloped in the 1980s, however the Victorian locomotive shed in the northern half of the site, which is a listed building, was kept in place and functions today as an exhibition space. The south-eastern half of the site is occupied by 5-storey office building from the 1980s, this building has a reinforced concrete frame. Both existing buildings must remain operational during the current development. The northern and southern flanks of the site are bounded by busy main roads. The Local Authority has set a number of traffic restrictions which limit construction traffic to times outside of morning and evening rush hour.

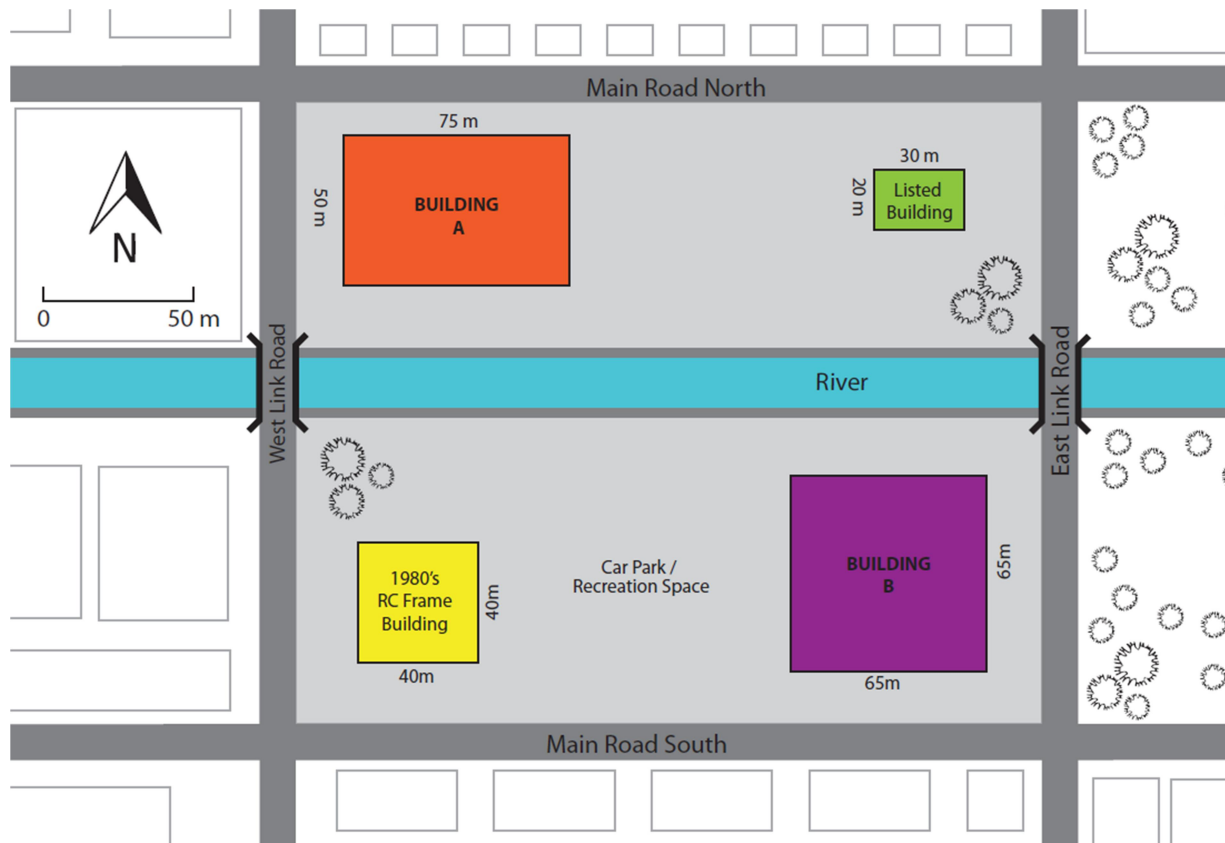


Figure 1: Layout of the Proposed Development

Building A is a 4-storey structure which will house artists' studios and creative workshops on the ground floor and office spaces on all other floors. Building B is a 6-storey structure with retail space on the ground floor and residential space on all other floors. In both buildings A and B, a planted green roof is to be incorporated for environmental sustainability. The client requires that all design options must take into account the presence of existing structures on and adjacent to the site and must minimise any possible disruption.

2.0 STRUCTURAL DESIGN BRIEF:

1. The outline floor plans and sections of buildings A and B are shown in Figure 2. In both buildings, the client wishes to have as much open plan space as possible, any columns or vertical elements must be no less than 7.0m apart.
2. For both buildings each storey is to have an overall height of 4.6m. The client requires that the ground floor has a clear height (floor to ceiling) of 3.75m, while the remaining floors require a clear height of 3.5m. The integration of services should be considered to ensure they can be accommodated and do not impede the clear floor height required.
3. The fire resistance is to be 1 hour.
4. The characteristic imposed loads are as follows:

Office Building (Building A):

Roof:	Imposed	7.5 kN/m ²		
1 st to 3 rd floors:	Imposed	4.0 kN/m ²	Services	0.25 kN/m ²
Ground floor:	Imposed	5.0 kN/m ²	Services	0.25 kN/m ²

Residential Building (Building B):

Roof:	Imposed	7.5 kN/m ²		
1 st to 5 th floors:	Imposed	3.0 kN/m ²	Services	0.25 kN/m ²
Ground floor:	Imposed	5.0 kN/m ²	Services	0.25 kN/m ²

Design Requirements

- a) Provide the client with two distinct floor options for the building; one option should be based on a concrete frame solution, the other based on a steel frame solution. Use hand sketches to show the layout and how loads would be transferred to the columns. Also indicate how lateral stability can be achieved (no calculations necessary)

(formative)

- b) From part a) appraise each option and identify the solution you recommend, giving reasons for your choices. Note that a qualitative appraisal of the carbon footprint should be among the various rationale included in the option appraisal.

(10 marks)

- c) For your chosen solution, complete a load take-down to determine column loads and forces in floor beams. Using this, produce preliminary sizing calculations for a primary beam, secondary beam (where applicable) and column. Calculations should be a maximum of 10 pages. The ULS partial factors of safety for dead and imposed loads are 1.35 and 1.5 respectively.

(15 marks)

- d) Using appropriate engineering scales, produce one or two A3 drawings for your chosen scheme that includes a typical floor plan, section, elevation and any important details.

(12 marks)

- e) Briefly describe the method of construction that could be used to build your design solution, indicating key information for the Contractor (maximum of 500 words).

(8 marks)

- f) The existing office building is a 5-storey in-situ reinforced concrete frame structure constructed in the 1980s. Discuss how this building might be refurbished and potential options to add an additional floor, include hand sketches to illustrate.

(5 marks)

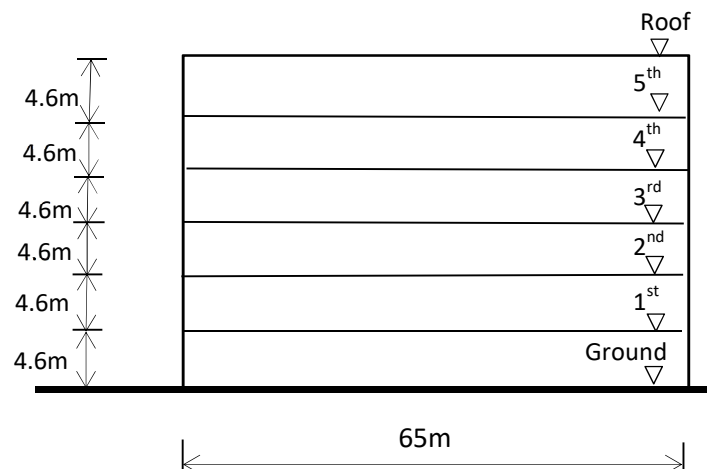
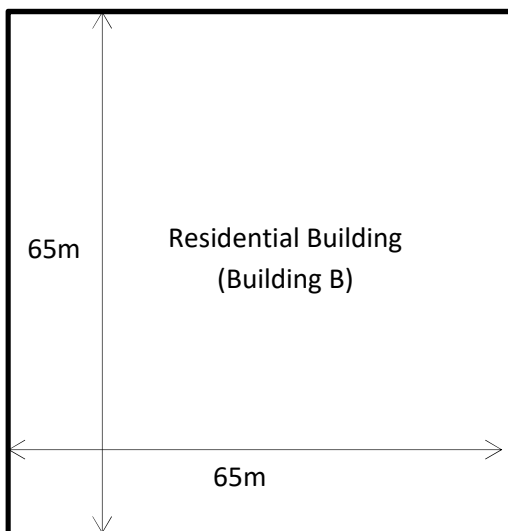
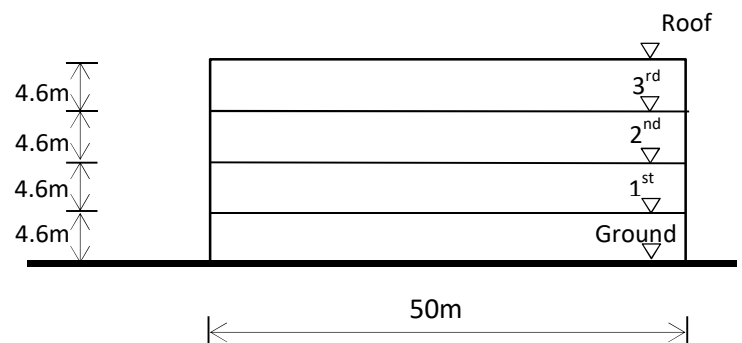
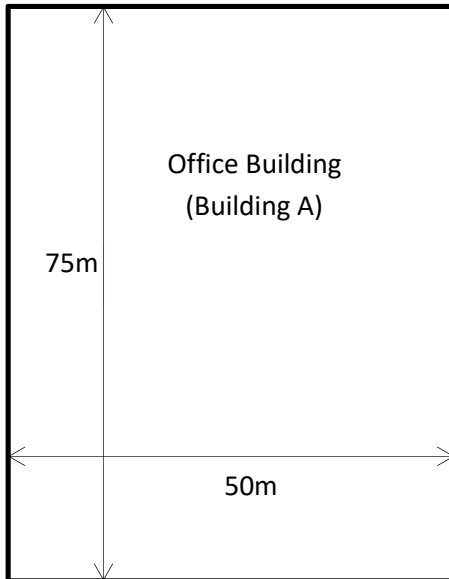


Figure 2: Building A(top), Building B (bottom), Plan views (left), Sections (right), NOT TO SCALE

3.0 HYDRAULICS DESIGN BRIEF

In this section you will consider the potential hydrological impacts of your design. This will cover aspects relating to drainage of the site and flood risk from the river. You will design two aspects of drainage related to peak design rainfall for your building in your assigned city: drainage from the roof of the building, and drainage of a car parking area, along with initial sizing of the drainage pipes to drain the site.

3.1 DRAINAGE STUDY

a) Initial task

Roof drainage: Draw a plan view of one of your buildings (and sections if necessary), showing the gutters and drainpipes for your building, clearly indicating which roof area drains into which gutters, and how those then drain into downpipes. You must design for a conventional impermeable roof but may also consider a green roof.

Car park: Devise an outline design for a car park to take at least 20 cars. A generous car parking space is 2.5 m wide by 5 m long, and a gap of 6 m is needed between rows of spaces (or spaces and boundaries) if the gap is to be used to manoeuvre into the space. Also include a roadway between the car park and a neighbouring road (but we won't be looking at the road drainage here). You will consider two options for drainage: simple drainage into gullies within the car park, or drainage initially via swales around or within the car park. Supply drawings for both options, clearly showing which areas of the car park drain into which gullies/swales. Again, you must design for an impermeable surface for the car park, but may also consider other options.

Pipe networks: Provide drawings showing how the rainwater from the roof and car park is taken off the site at any boundary (other than the river).

(formative)

b) Roof drainage

Design a simple drainage system based on semi-circular gutters for the roof of your building. Specify the size of gutters, downpipes and pipe network required to transport water from the roof, and calculate peak flows in each part of your design. Explain the basis of your design on rainfall estimates, return periods and concentrations times. Comment on the impact a green roof would have on the drainage system required.

(10 marks)

c) Car Park drainage

Design a drainage system for your car park. Specify the size of gullies or swales and the pipe network required and calculate peak flows in each part of the system. Explain the basis of your design on rainfall

estimates, return periods and concentrations times. Comment on the impact SuDS would have on the drainage system required.

(10 marks)

3.2 FLOOD STUDY

You will perform a flood risk analysis of the area illustrated in Figure 3 using the software HEC-RAS. The geometry of the river cross-sections and the distances between the sections are provided in a separate Excel file available on Blackboard in Hydraulics>Flooding>Coursework Data. Each group has been assigned a different data file named “river-data-#n.xls”, where #n is the group number. Details of the bridge located immediately downstream of section S-7 and the broad crested weir downstream of section S-1 are given in Figures 4 and 5, respectively. The other bridges along the river do not affect the flow and therefore will not be considered in this analysis. The average bed slope is 0.0005 and the channel is mostly straight with negligible vegetation. The annual peak flow discharges measured at an upstream gauge station from the year 1967 to 2016 are given in the same Excel file.

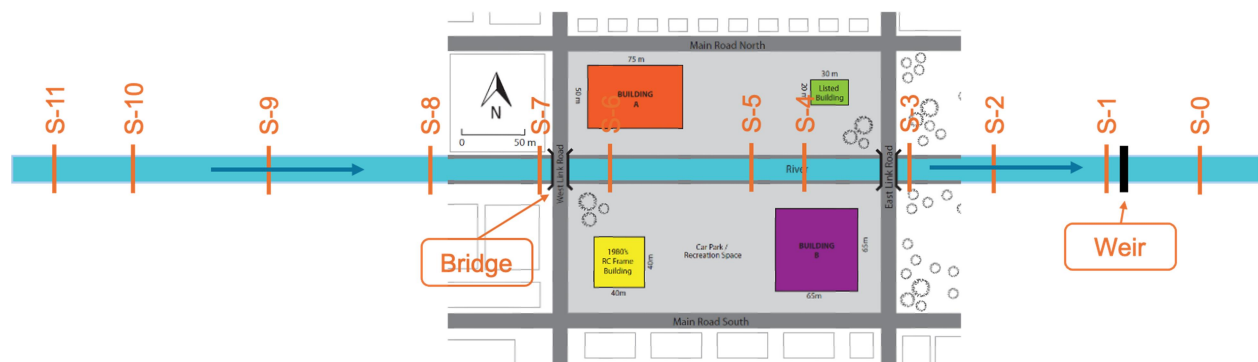


Figure 3. Site map

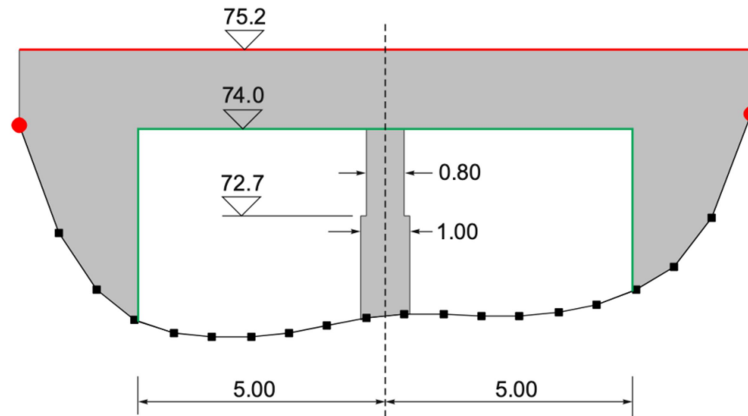


Figure 4. Geometry of the bridge downstream of section S-7. Dimensions are in metres and you can treat this section as perpendicular to the river. The width of the bridge is 7.5 m, and the bridge pier is square nosed.

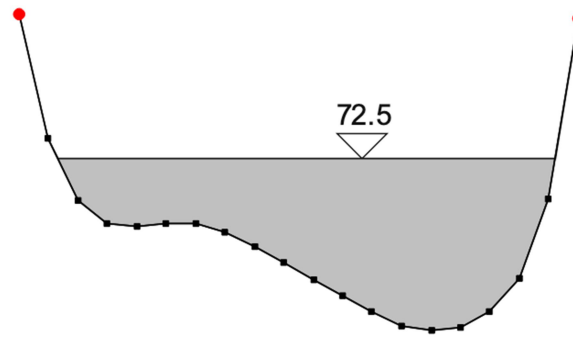


Figure 5. Geometry of the broad crested weir downstream of section S-1. Dimensions are in metres. The width of the weir in the direction of the flow is 15 m.

(a) Perform a flood frequency analysis and determine the flood discharge corresponding to 10, 50 and 100-year return periods.

(7 marks)

(b) For the same (steady) discharges, determine the river water surface profiles using HEC-RAS. Discuss the sensitivity of the model results to any assumptions you have made.

(16 marks)

(c) Discuss the risk of flooding in the area under the three scenarios above and suggest potential remedies. Comment on whether the removal of the weir and/or the modification of the bridge could help alleviate the risk of flooding.

(7 marks)

4.0 GEOTECHNICAL DESIGN BRIEF:

The aim of the project is to explore suitable concepts and options for foundations of Buildings A and B based on the results of a preliminary site investigation and in-situ tests. The results of Cone Penetration Test (CPT), which is a well-established in-situ test, are to be used for the design of suitable foundation concepts for these two buildings. It is noted that the CPT profile of the soil is provided as a separate document on the blackboard.

Below are the requirements of this design work:

- a) To calculate the bearing capacity of shallow foundations from the direct and indirect use of CPT results.

(10 marks)

- b) To provide an estimation of the immediate settlement of shallow foundations based for a reasonable range of foundation widths.

(8 marks)

- c) To provide the calculation and summary of the bearing capacity of single pile foundation through the direct use of CPT results.

(10 marks)

- d) From the results of tasks A to C, to present recommendations on a suitable option for typical foundation required for both buildings A and B (separately) based on the maximum structure load. The recommendations should at least include the type of foundation and accordingly further site investigations for the detailed design phase.

(7 marks)

- e) Provide further recommendations and preliminary analysis of group pile foundations if single pile is not adequate for the foundation.

(5 marks)

4.1. DESIGN ASSUMPTIONS

1. Only square footings (spread footing) are considered for shallow foundation options.
2. Shallow foundations depth is 0.8 m depth below ground surface.
3. Immediate settlement of shallow foundation options is required to be limited to 25 mm.

4. The spread footing widths are between 1.0 m and 4.0m, subject to the condition that the individual footing only supports a single structural column.
5. Only plain bored concrete pile foundations are to be considered.
6. The diameter for the concrete piles is 300 mm.
7. Calculation of the pile foundation settlement is not required.
8. Length of the pile foundation should be >25m.

5.0 GENERAL SUBMISSION REQUIREMENTS:

- Each submission should consist of a single pdf file submitted to the unit's Blackboard site.
- Each submission should include a completed individual mark weighting table (table is available on the unit's Blackboard site).